# ShopEZ: E-commerce Application - Report

## 1. Introduction

**Project Title:** ShopEZ: E-commerce Application

**Team Members:**

- C Chaithanya Prasad
- T Harshitha
- Yaswanth Palapati

## 2. Project Overview

**Purpose:**
ShopEZ is designed to cater to the growing need for efficient and user-friendly e-commerce platforms. It bridges the gap between complex online shopping systems and user convenience by providing streamlined navigation, secure transactions, and personalized product recommendations.Additionally,theplatformempowerssellerswithadashboardformanaging their inventory, processing orders, and accessing analytics to drive growth.

**Features:**

1. **SeamlessCheckout:** Secure and smooth payment process with instant order confirmations and email notifications.
2. **EffortlessProductDiscovery:** Advanced search capabilities, intuitive category navigation, and powerful filters to help users find exactly what they need.
3. **PersonalizedRecommendations:** AI-driven algorithms analyze user behavior to provide curated product suggestions.
4. **SellerDashboard:** Comprehensive tools for inventory tracking, order processing, and analytics to monitor performance metrics.
5. **Real-timeAnalytics:** Data-driven insights for sellers, highlighting sales trends, customer preferences, and product performance.

## 3. Architecture

**Frontend:**

- Developedusing**React.js**foritscomponent-basedarchitectureandstatemanagement capabilities.
- Featuresdynamiccomponentslike:
  - **ProductListings**:Displaysproductswithsortingandfilteringoptions.
  - **CartManagement**:Allowsuserstoadd,update,orremoveitemsintheircart.
  - **UserAuthentication**:Loginandregistrationpageswithsecurevalidation.
  - **AdminPanel**:Providessellerswithtoolstomanageinventoryandview analytics.

**Backend:**

- Built with **Node.js** and **Express.js**, ensuring scalability and high performance.
- Features include:
  - **APIEndpoints**:
    - /productsforfetchingproductdata.
    - /ordersforprocessingcustomerorders.
    - /usersformanaginguserauthentication.
  - **Middleware** for error handling and authentication using **JWT**.

**Database:**

- **MongoDB** serves as the database, storing collections for:
  - **Users**:Authenticationcredentials,profiles,andpurchasehistory.
  - **Products**:Informationoninventory,prices,categories,anddescriptions.
  - **Orders**:Detailsaboutplacedorders,deliverystatus,andpayment.

This architecture ensures modularity, scalability, and efficient data management.

## 4. Setup Instructions

**Prerequisites:**

- **Node.js** : v14 or later
- **MongoDB** : Installed locally or set up using a cloud provider like MongoDB Atlas
- **npm**: Package manager for installing dependencies

**Installation: Clone the repository:**

`https://github.com/Chaithanya182/shopez.git`

**Navigate to the project directory:**

```
cd shopEZ
```

**Install dependencies for the backend and frontend:**

Backend:

```
cd server
npm install
```

Frontend:

```
cd client
npm install
```

**Run the servers:**

Backend:

```
node index.js
```

Frontend:

```
npm start
```

## 5. Folder Structure

**Client:**

- src/components: Contains reusable components like **Navbar**, **ProductCard**, **CartItem**, etc.
- src/pages: Holds page components such as **Home**, **Cart**, **Checkout**, and **AdminPanel**.
- src/services: Manages API interactions for fetching and posting data.
- src/redux: Implements state management for cart items, user authentication, and order status.

**Server:**

- `routes`: Defines all RESTful API routes for users, products, and orders.
- `controllers`: Contains the logic for handling API requests and responses.
- `models`: Defines database schemas for **Users**, **Products**, and **Orders**.
- `middleware`: Handles authentication (JWT) and error management.

## 6. Running the Application

**Frontend:**

Navigate to the client directory:

```
cd client
```

Start the React application:

```
npm start
```

**Backend:**

Navigate to the server directory:

```
cd server
```

Start the Node.js server:

```
node index.js
```

## 7. API Documentation

**Endpoints:**

1. **GET/products**
   - Fetches the list of all products.
   - Parameters: Optional category and price filters.

Response:

json

```
[
  {
    "id": "123",
    "name": "Gold Bracelet",
    "price": 50,
    "category": "Accessories"
  }
]
```

2. **POST /orders**
   - Places a new order.
   - Parameters: User ID, product details, and quantity.

Response:
json

```
{
  "message": "Order placed successfully",
  "orderId": "456"
}
```

## 8. Authentication

**JWT-based Authentication:**

- **Login** : Issues a JWT token upon successful authentication.
- **TokenValidation** : Protects private routes like /orders and `/admin`.
- **Logout** : Invalidates the token on the client side.

## 9. User Interface

**Screens:**

- **Home:** Displays trending products and categories.
- **ProductDetails:** Shows detailed information about a selected item.
- **Cart:** Summarizes selected products and their quantities.
- **AdminDashboard:** Offers order status updates and analytics for sellers.

## 10. Testing

- **Unit Testing:** Conducted using **Jest** for components and backend logic.
- **API Testing:** Performed using **Postman** to validate endpoints.

## 11. Screenshots or Demo

Screenshots of key UI components and admin panel.

File Edit Selection View Go Run Terminal Help shopEZ-nm

EXPLORER

SHOPEZ...
- client
- node_modules
- server
  - node_modules
  - index.js
  - package-lock.json
  - package.json
  - Schema.js
- package-lock.json
- package.json
- README.md

OUTLINE
TIMELINE

Schema.js

server > Schema.js > ...

```
14
15  const productSchema = new mongoose.Schema({
16      title: {type: String},
17      description: {type: String},
18      mainImg: {type: String},
19      carousel: {type: Array},
20      sizes: {type: Array},
21      category: {type: String},
22      gender: {type: String},
23      price: {type: Number},
24      discount: {type: Number}
25  })
26
27  const orderSchema = new mongoose.Schema({
28      userId: {type: String},
29      name: {type: String},
30      email: {type: String},
31      mobile: {type: String},
32      address: {type: String},
33      pincode: {type: String},
34      title: {type: String},
35      description: {type: String},
36      mainImg: {type: String},
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\HP\OneDrive\Desktop\shopEZ-nm\server> npm install

up to date, audited 160 packages in 7s

17 packages are looking for funding
  run `npm fund` for details

found 0 vulnerabilities
PS C:\Users\HP\OneDrive\Desktop\shopEZ-nm\server> node index.js
running @ 6001
```

MarsCode AI Ln 1, Col 1 Spaces: 4 UTF-8 LF {} JavaScript

File Edit Selection View Go Run Terminal Help shopEZ-nm

EXPLORER

SHOPEZ...
- client
- node_modules
- server
  - node_modules
  - index.js
  - package-lock.json
  - package.json
  - Schema.js
- package-lock.json
- package.json
- README.md

OUTLINE
TIMELINE

index.js

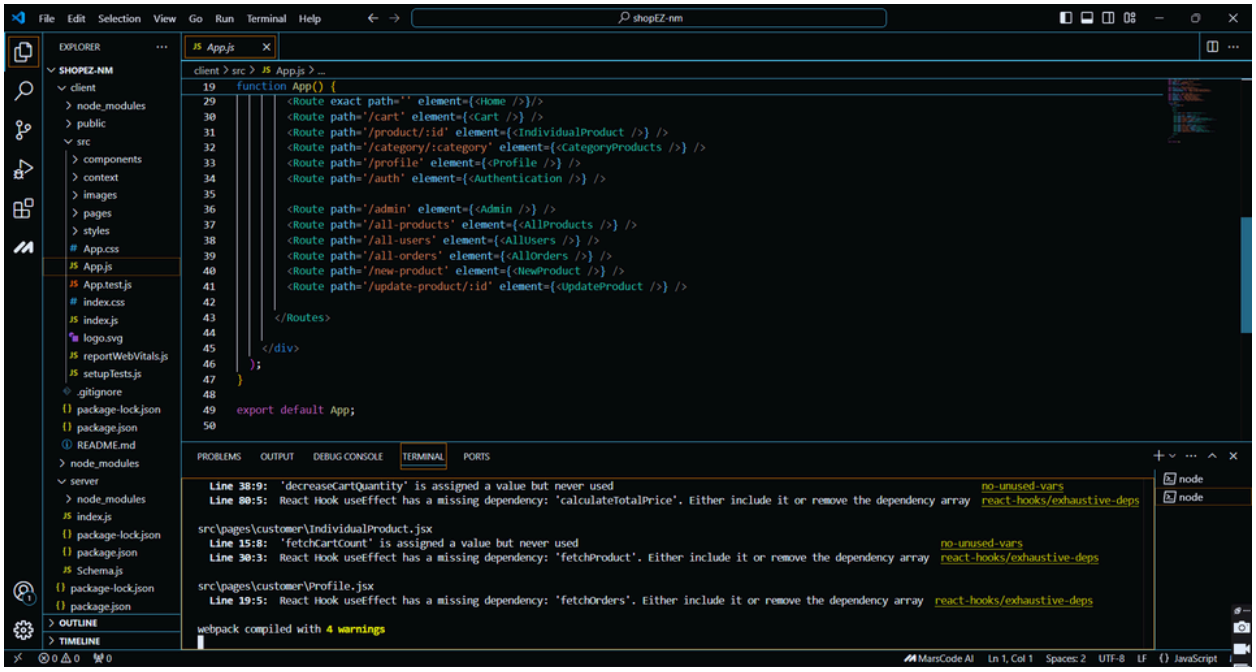
server > index.js > ...

```
20   }).then(()=>{
337      app.put('/decrease-cart-quantity', async(req, res)=>{
343
344          res.json({message: 'decremented'});
345      }catch(err){
346          res.status(500).json({message: "Error occured"});
347      }
348  })
349
350
351  // remove from cart
352
353  app.put('/remove-item', async(req, res)=>{
354      const {id} = req.body;
355      try{
356          const item = await Cart.deleteOne({_id: id});
357          res.json({message: 'item removed'});
358      }catch(err){
359          res.status(500).json({message: "Error occured"});
360      }
361  });
362
363
364  // Order from cart
```

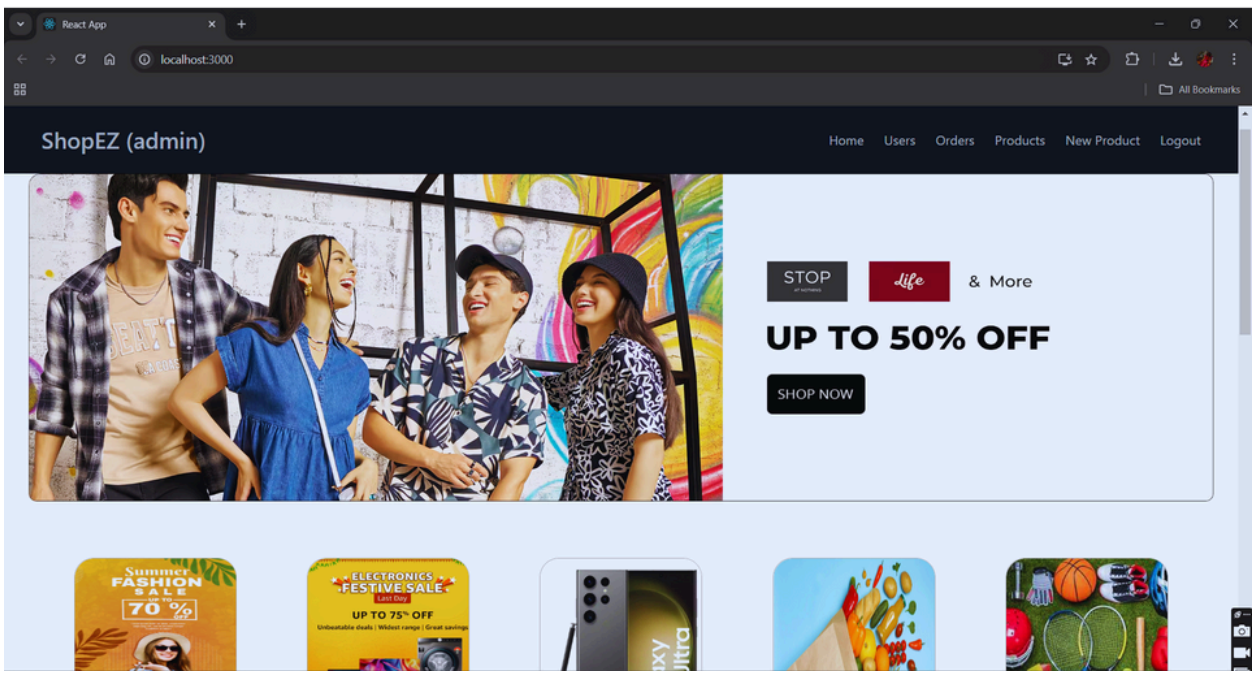
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

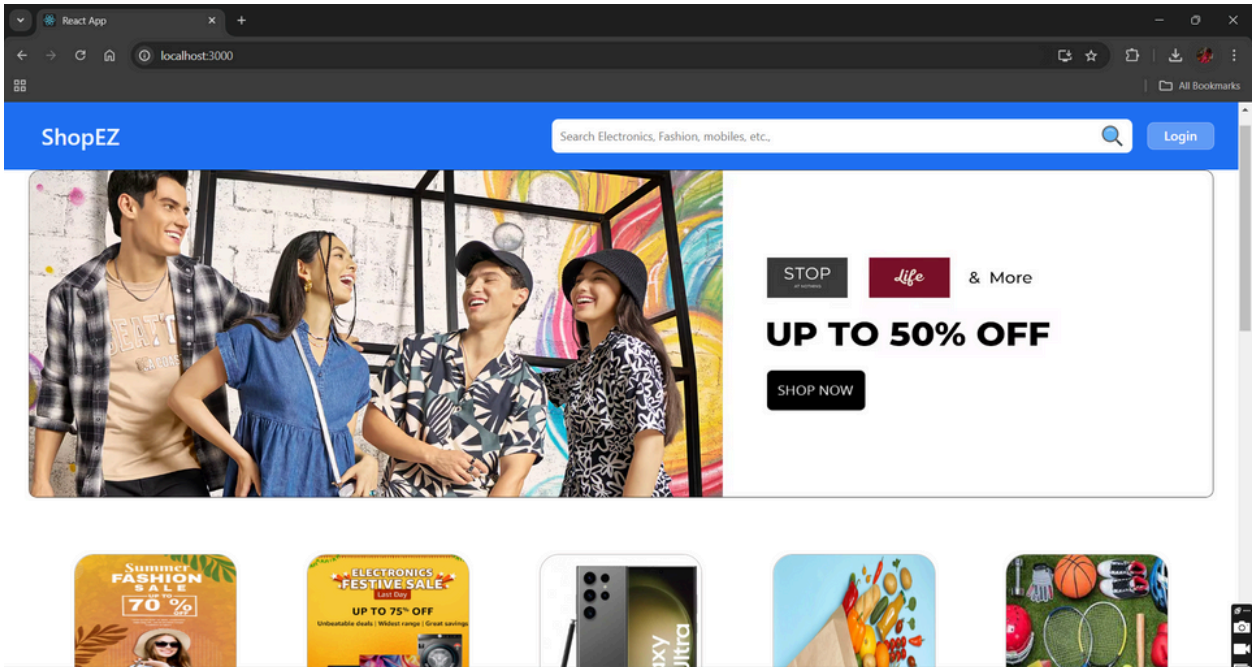
```
PS C:\Users\HP\OneDrive\Desktop\shopEZ-nm\server> npm install

up to date, audited 160 packages in 7s

17 packages are looking for funding
  run `npm fund` for details

found 0 vulnerabilities
PS C:\Users\HP\OneDrive\Desktop\shopEZ-nm\server> node index.js
running @ 6001
```

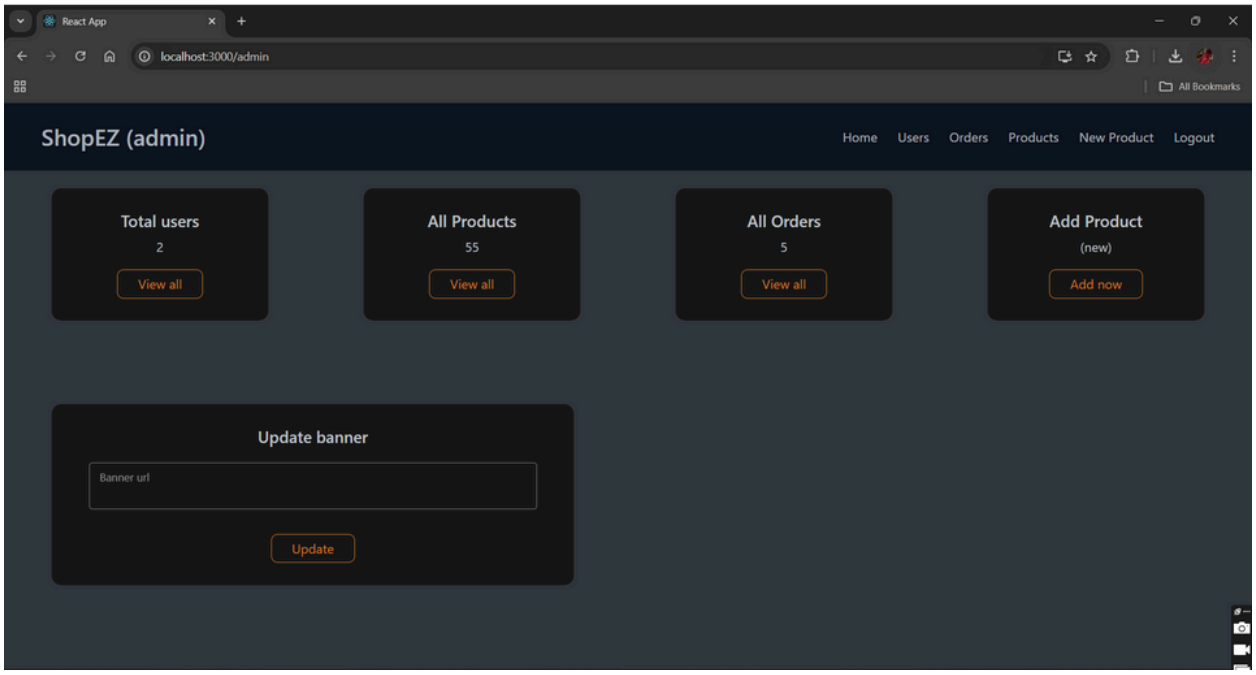
MarsCode AI Ln 1, Col 1 Spaces: 4 UTF-8 LF {} JavaScript

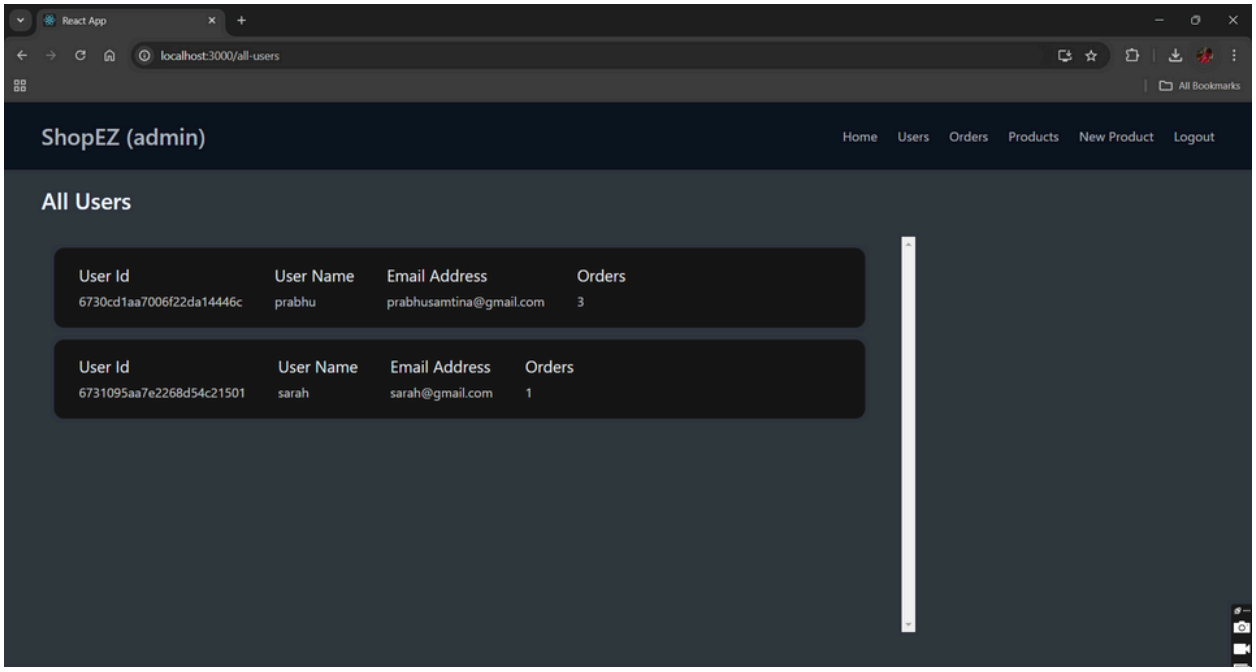

File Edit Selection View Go Run Terminal Help
shopEZ-nm
EXPLORER
SHOPEZ-NM
client
node_modules
public
src
components
context
images
pages
styles
App.css
App.js
App.test.js
index.css
index.js
logo.svg
reportWebVitals.js
setupTests.js
.gitignore
package-lock.json
package.json
README.md
node_modules
server
node_modules
index.js
package-lock.json
package.json
Schema.js
package-lock.json
package.json
OUTLINE
TIMELINE
client > src > App.js > ...
function App() {
<Route exact path='' element={<Home />}/>
<Route path='/cart' element={<Cart />} />
<Route path='/product/:id' element={<IndividualProduct />} />
<Route path='/category/:category' element={<CategoryProducts />} />
<Route path='/profile' element={<Profile />} />
<Route path='/auth' element={<Authentication />} />
<Route path='/admin' element={<Admin />} />
<Route path='/all-products' element={<AllProducts />} />
<Route path='/all-users' element={<AllUsers />} />
<Route path='/all-orders' element={<AllOrders />} />
<Route path='/new-product' element={<NewProduct />} />
<Route path='/update-product/:id' element={<UpdateProduct />} />
</Routes>

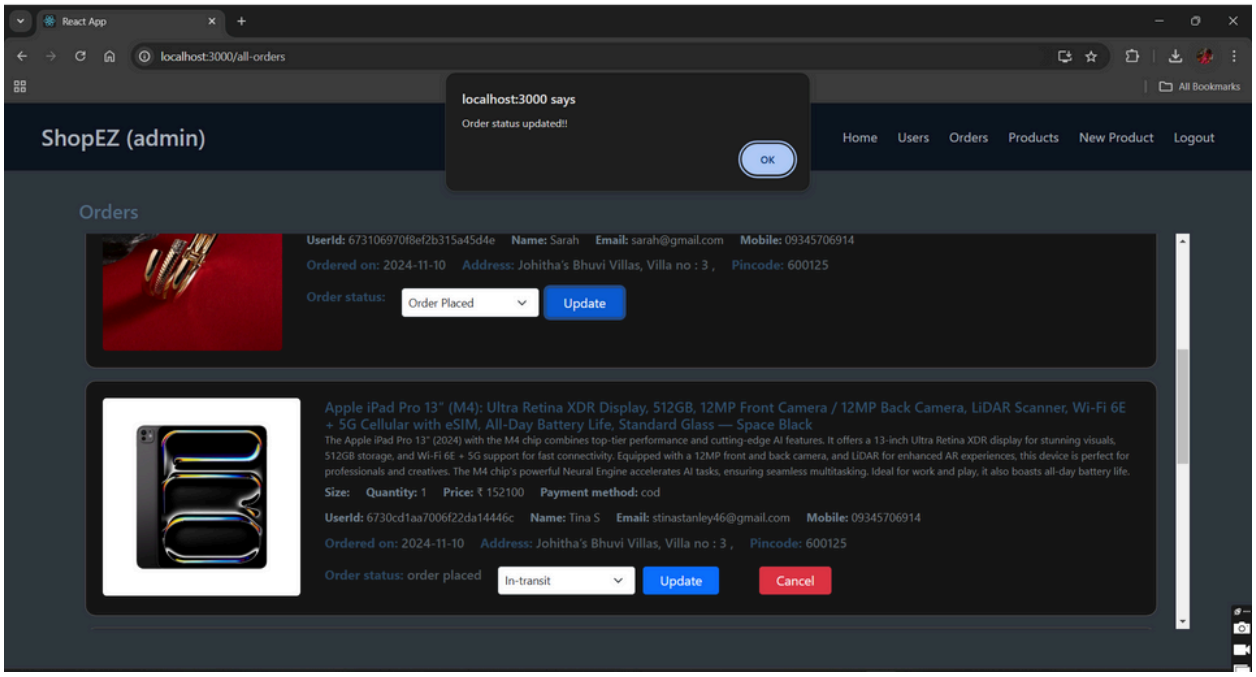
);
}
export default App;
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
Line 38:9: 'decreaseCartQuantity' is assigned a value but never used no-unused-vars
Line 80:5: React Hook useEffect has a missing dependency: 'calculateTotalPrice'. Either include it or remove the dependency array react-hooks/exhaustive-deps
src\pages\customer\IndividualProduct.jsx
Line 15:8: 'fetchCartCount' is assigned a value but never used no-unused-vars
Line 30:3: React Hook useEffect has a missing dependency: 'fetchProduct'. Either include it or remove the dependency array react-hooks/exhaustive-deps
src\pages\customer\Profile.jsx
Line 19:5: React Hook useEffect has a missing dependency: 'fetchOrders'. Either include it or remove the dependency array react-hooks/exhaustive-deps
webpack compiled with 4 warnings
node
node
MarsCode AI Ln 1, Col 1 Spaces: 2 UTF-8 LF JavaScript

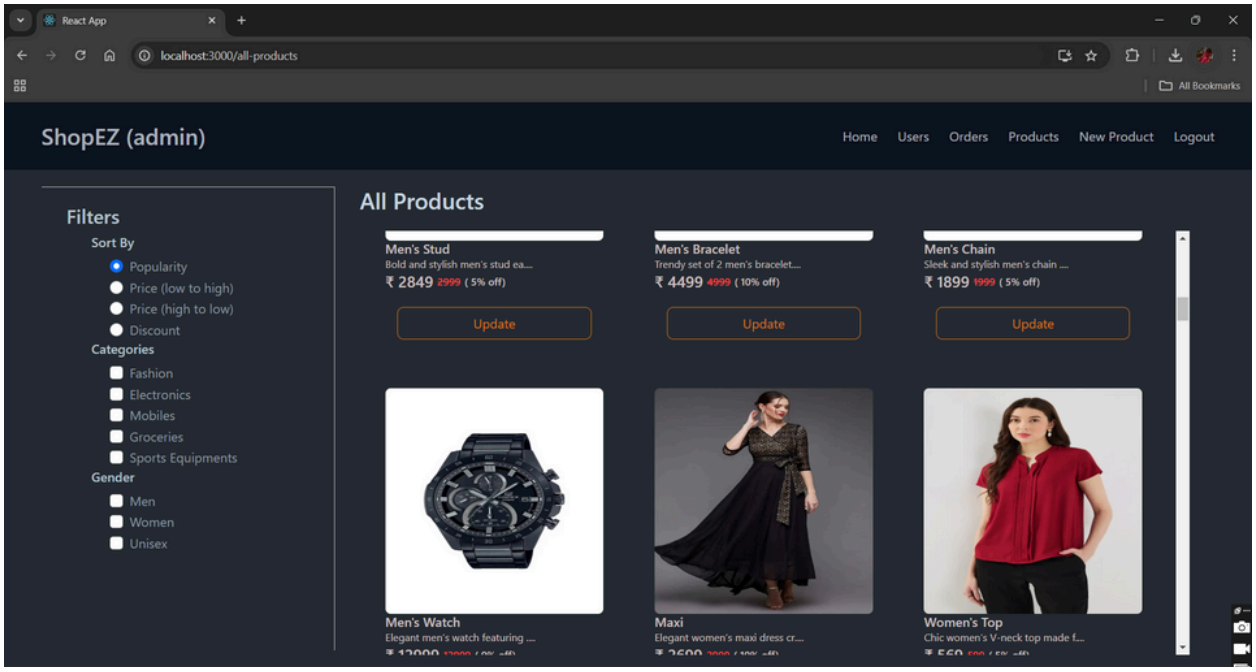

React App
localhost:3000
All Bookmarks
ShopEZ (admin)
Home Users Orders Products New Product Logout
STOP
life
& More
UP TO 50% OFF
SHOP NOW
Summer FASHION SALE
70 %
ELECTRONICS FESTIVE SALE
UP TO 75% OFF

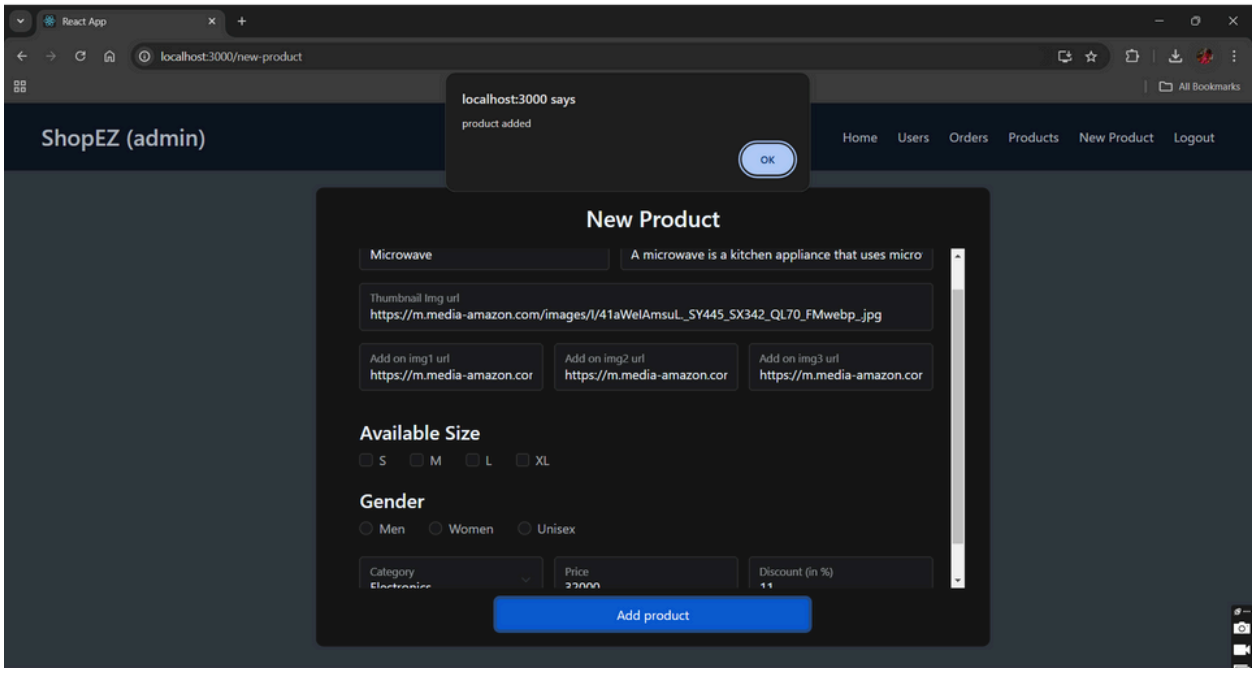

React App
localhost:3000
All Bookmarks
ShopEZ
Search Electronics, Fashion, mobiles, etc.,
Login
STOP
life
& More
UP TO 50% OFF
SHOP NOW
Summer FASHION SALE
70 %
ELECTRONICS FESTIVE SALE
UP TO 75% OFF

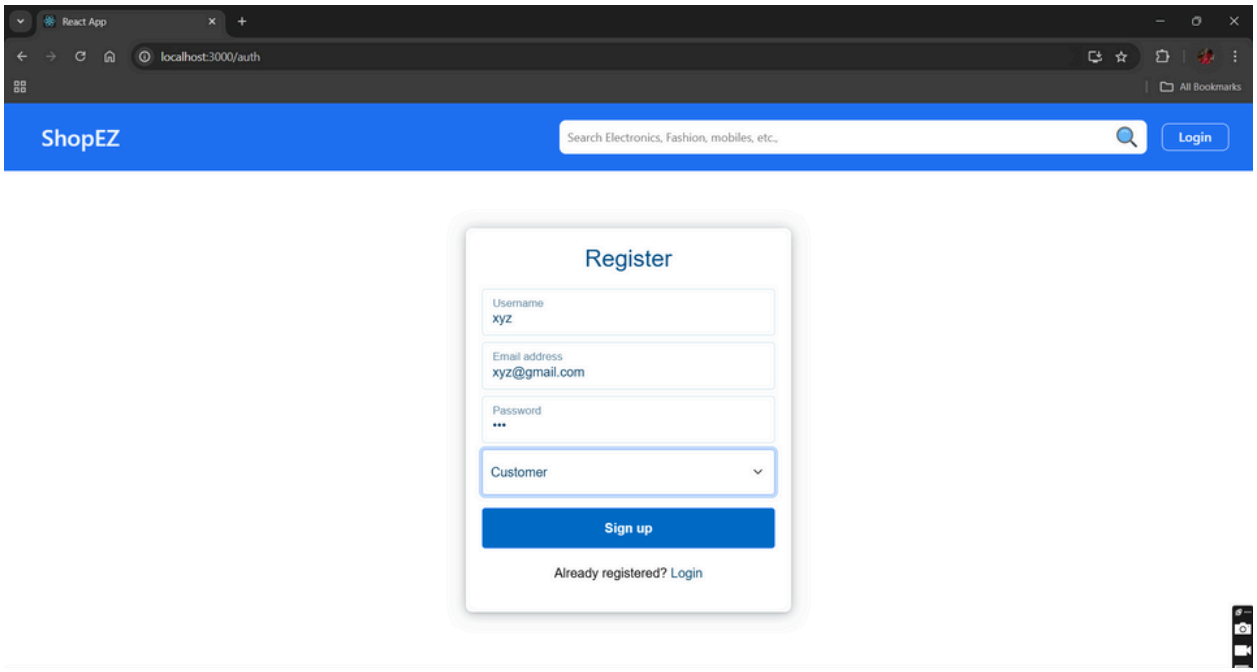

React App
localhost:3000/admin
All Bookmarks
ShopEZ (admin)
Home
Users
Orders
Products
New Product
Logout
Total users
2
View all
All Products
55
View all
All Orders
5
View all
Add Product
(new)
Add now
Update banner
Banner url
Update

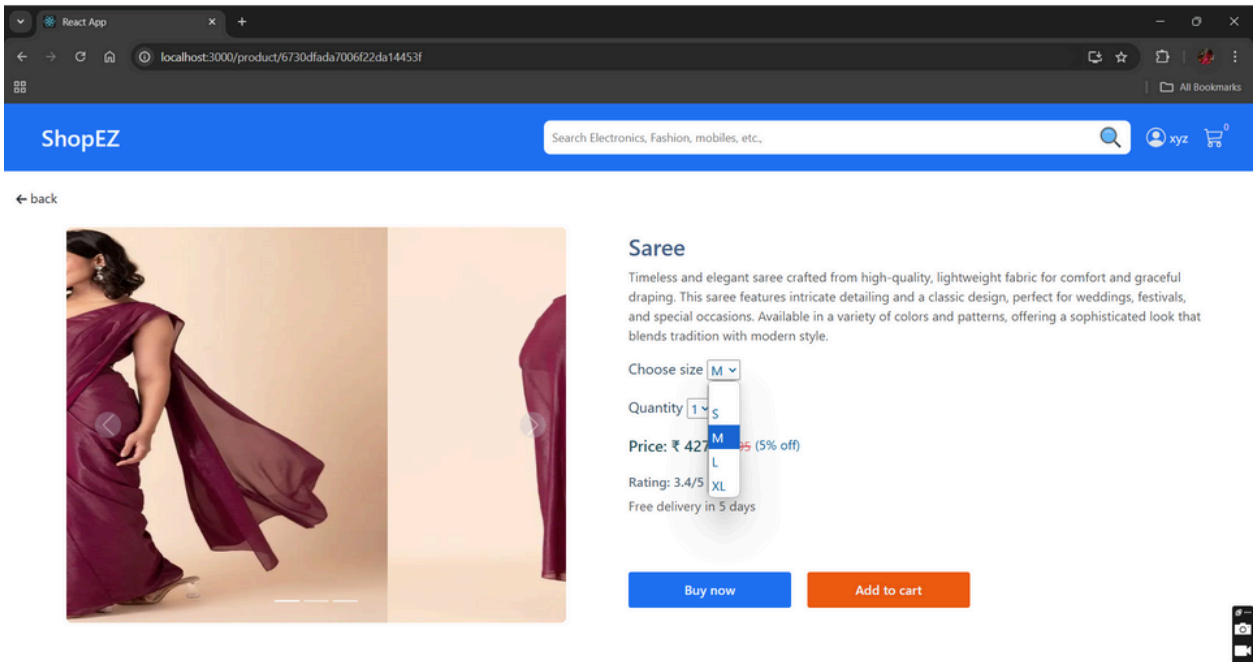

React App
localhost:3000/all-users
All Bookmarks
ShopEZ (admin)
Home
Users
Orders
Products
New Product
Logout
All Users
User Id
User Name
Email Address
Orders
6730cd1aa7006f22da14446c
prabhu
prabhusamtina@gmail.com
3
User Id
User Name
Email Address
Orders
6731095aa7e2268d54c21501
sarah
sarah@gmail.com
1

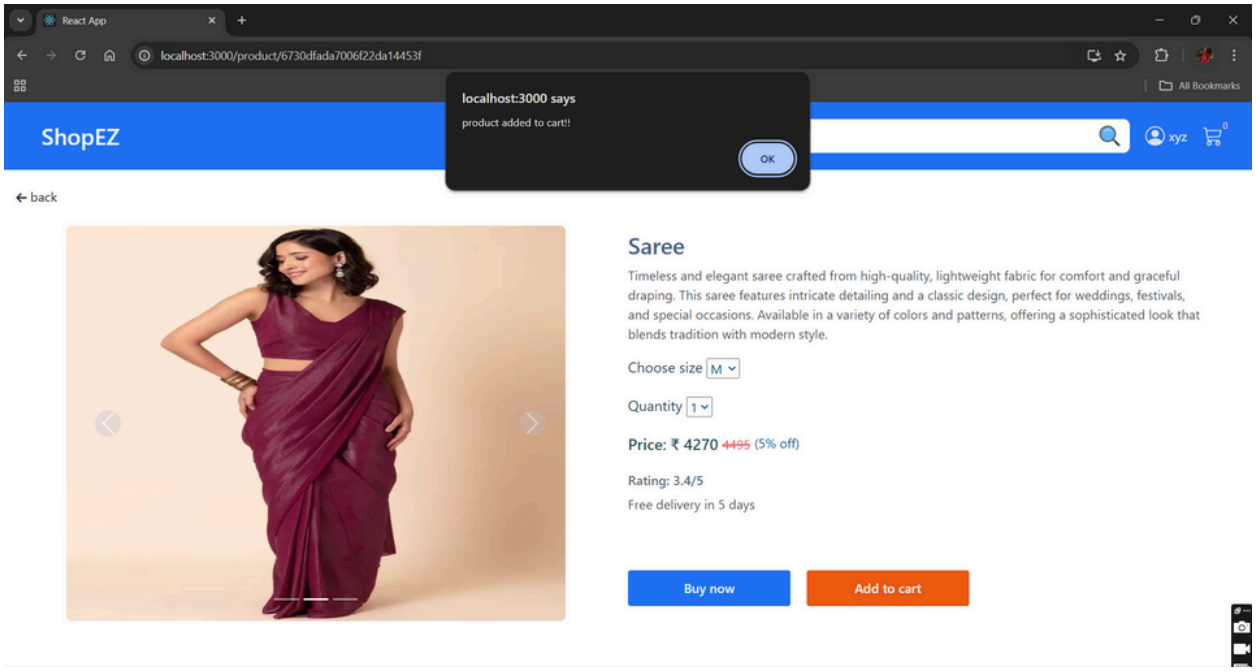

React App
localhost:3000/all-orders
All Bookmarks
localhost:3000 says
Order status updated!!
OK
ShopEZ (admin)
Home
Users
Orders
Products
New Product
Logout
Orders
UserId: 673106970f8ef2b315a45d4e
Name: Sarah
Email: sarah@gmail.com
Mobile: 09345706914
Ordered on: 2024-11-10
Address: Johitha's Bhuvi Villas, Villa no : 3 ,
Pincode: 600125
Order status:
Order Placed
Update
Apple iPad Pro 13" (M4): Ultra Retina XDR Display, 512GB, 12MP Front Camera / 12MP Back Camera, LiDAR Scanner, Wi-Fi 6E + 5G Cellular with eSIM, All-Day Battery Life, Standard Glass — Space Black
The Apple iPad Pro 13" (2024) with the M4 chip combines top-tier performance and cutting-edge AI features. It offers a 13-inch Ultra Retina XDR display for stunning visuals, 512GB storage, and Wi-Fi 6E + 5G support for fast connectivity. Equipped with a 12MP front and back camera, and LiDAR for enhanced AR experiences, this device is perfect for professionals and creatives. The M4 chip's powerful Neural Engine accelerates AI tasks, ensuring seamless multitasking. Ideal for work and play, it also boasts all-day battery life.
Size:
Quantity: 1
Price: ₹ 152100
Payment method: cod
UserId: 6730cd1aa7006f22da14446c
Name: Tina S
Email: stinastanley46@gmail.com
Mobile: 09345706914
Ordered on: 2024-11-10
Address: Johitha's Bhuvi Villas, Villa no : 3 ,
Pincode: 600125
Order status: order placed
In-transit
Update
Cancel

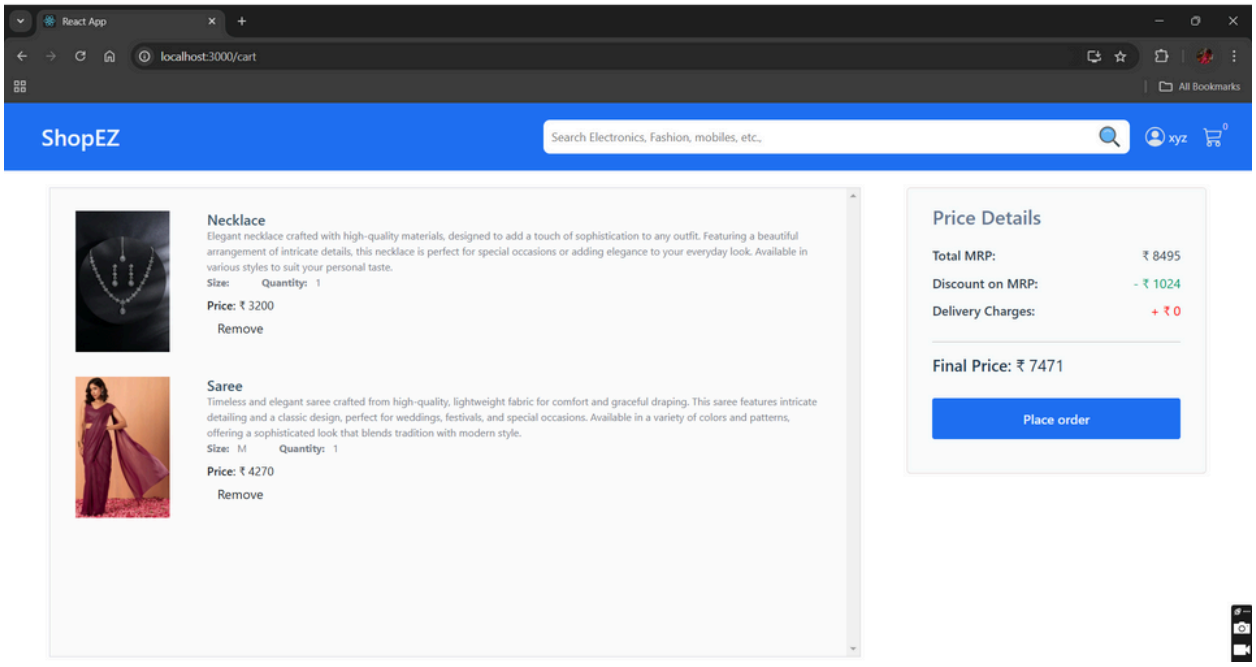

React App
localhost:3000/all-products
All Bookmarks
ShopEZ (admin)
Home
Users
Orders
Products
New Product
Logout
Filters
Sort By
Popularity
Price (low to high)
Price (high to low)
Discount
Categories
Fashion
Electronics
Mobiles
Groceries
Sports Equipments
Gender
Men
Women
Unisex
All Products
Men's Stud
Bold and stylish men's stud ea....
₹ 2849 2999 ( 5% off)
Update
Men's Bracelet
Trendy set of 2 men's bracelet....
₹ 4499 4999 ( 10% off)
Update
Men's Chain
Sleek and stylish men's chain ....
₹ 1899 1999 ( 5% off)
Update
Men's Watch
Elegant men's watch featuring ....
Maxi
Elegant women's maxi dress cr....
Women's Top
Chic women's V-neck top made f....

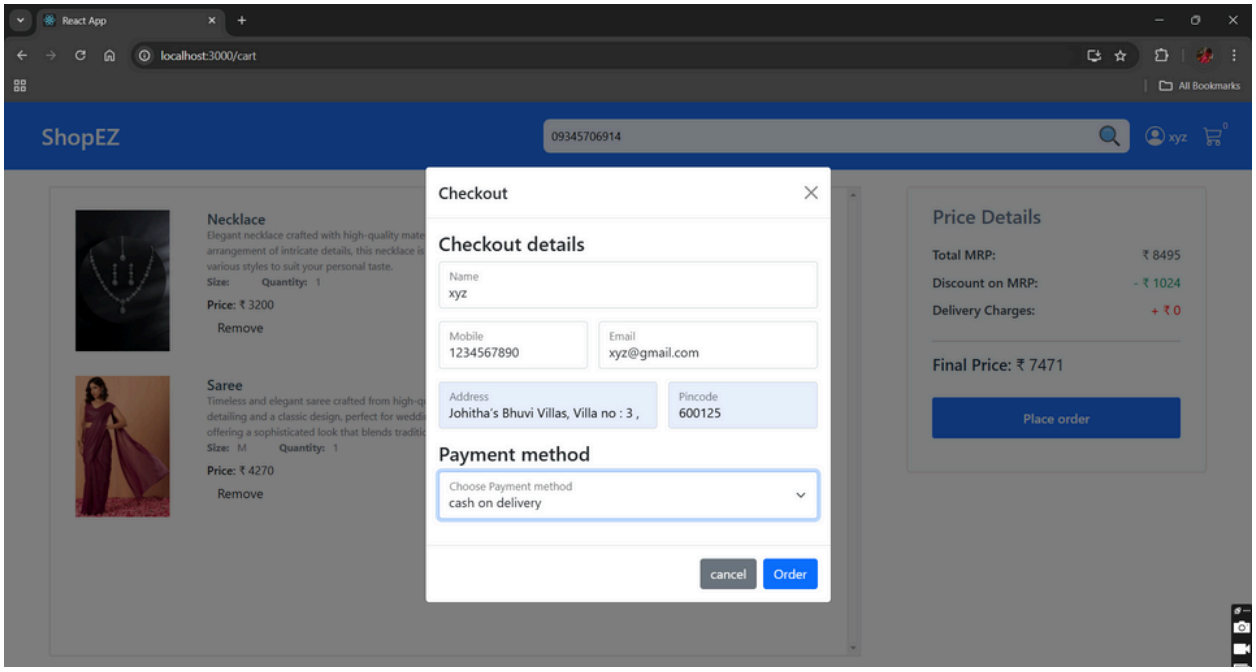

React App
localhost:3000/new-product
All Bookmarks
localhost:3000 says
product added
OK
ShopEZ (admin)
Home
Users
Orders
Products
New Product
Logout
New Product
Microwave
A microwave is a kitchen appliance that uses micro
Thumbnail Img url
https://m.media-amazon.com/images/I/41aWelAmsuL._SY445_SX342_QL70_FMwebp_.jpg
Add on img1 url
https://m.media-amazon.cor
Add on img2 url
https://m.media-amazon.cor
Add on img3 url
https://m.media-amazon.cor
Available Size
S
M
L
XL
Gender
Men
Women
Unisex
Category
Price
Discount (in %)
Add product

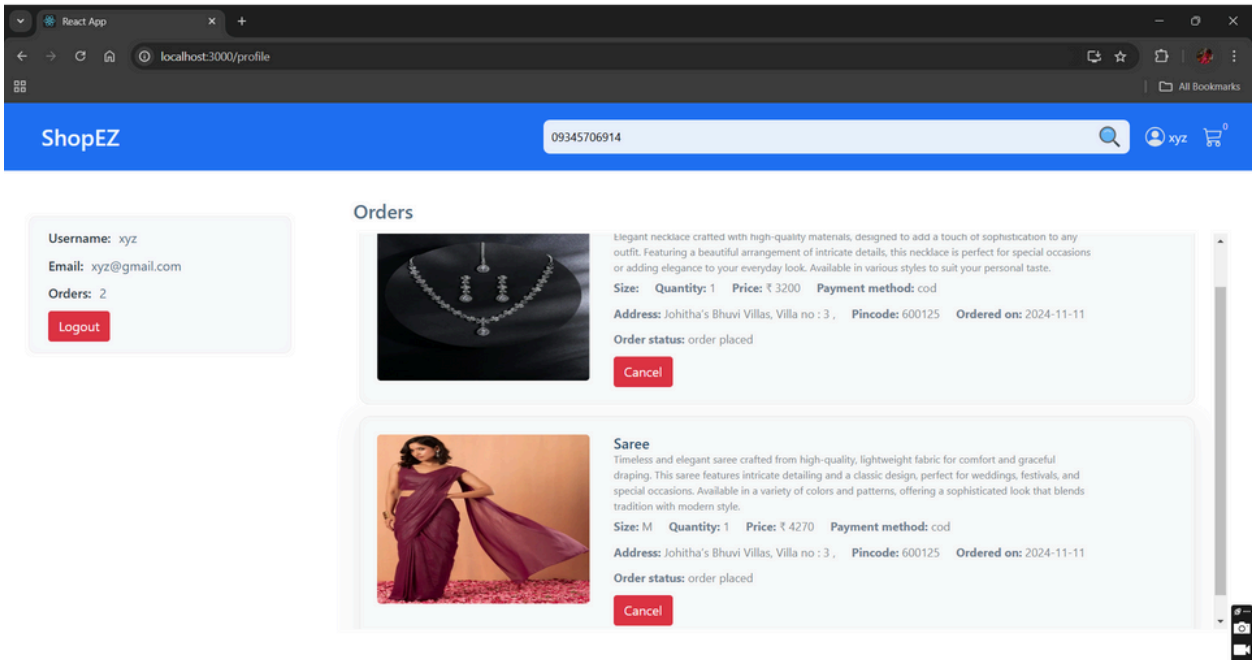

React App
localhost:3000/auth
All Bookmarks
ShopEZ
Search Electronics, Fashion, mobiles, etc.,
Login
Register
Username
xyz
Email address
xyz@gmail.com
Password
•••
Customer
Sign up
Already registered? Login


React App
localhost:3000/product/6730dfada7006f22da14453f
All Bookmarks
ShopEZ
Search Electronics, Fashion, mobiles, etc.,
xyz
back
Saree
Timeless and elegant saree crafted from high-quality, lightweight fabric for comfort and graceful draping. This saree features intricate detailing and a classic design, perfect for weddings, festivals, and special occasions. Available in a variety of colors and patterns, offering a sophisticated look that blends tradition with modern style.
Choose size M
S
M
L
XL
Quantity 1
Price: ₹ 427
(5% off)
Rating: 3.4/5
Free delivery in 5 days
Buy now
Add to cart


React App
localhost:3000/product/6730dfada7006f22da14453f
All Bookmarks
localhost:3000 says
product added to cart!!
OK
ShopEZ
xyz
back
Saree
Timeless and elegant saree crafted from high-quality, lightweight fabric for comfort and graceful draping. This saree features intricate detailing and a classic design, perfect for weddings, festivals, and special occasions. Available in a variety of colors and patterns, offering a sophisticated look that blends tradition with modern style.
Choose size M
Quantity 1
Price: ₹ 4270 4495 (5% off)
Rating: 3.4/5
Free delivery in 5 days
Buy now
Add to cart


React App
localhost:3000/cart
All Bookmarks
ShopEZ
Search Electronics, Fashion, mobiles, etc.,
xyz
Necklace
Elegant necklace crafted with high-quality materials, designed to add a touch of sophistication to any outfit. Featuring a beautiful arrangement of intricate details, this necklace is perfect for special occasions or adding elegance to your everyday look. Available in various styles to suit your personal taste.
Size: Quantity: 1
Price: ₹ 3200
Remove
Saree
Timeless and elegant saree crafted from high-quality, lightweight fabric for comfort and graceful draping. This saree features intricate detailing and a classic design, perfect for weddings, festivals, and special occasions. Available in a variety of colors and patterns, offering a sophisticated look that blends tradition with modern style.
Size: M Quantity: 1
Price: ₹ 4270
Remove
Price Details
Total MRP: ₹ 8495
Discount on MRP: - ₹ 1024
Delivery Charges: + ₹ 0
Final Price: ₹ 7471
Place order




Link to a live demo.

**Demo Video.mp4/**
**https://drive.google.com/file/d/1B025A59a0S6EKPuVxcbPbKBkCITjvwuN/view?usp=sharing**

## 12. Known Issues

- **Slow Search Performance:** Optimization is required for large product datasets.
- **UI Bugs:** Minor alignment issues on smaller screens.

## 13. Future Enhancements

- **Voice Search:** Enable users to search using voice commands.
- **Mobile App:** Create a cross-platform app using React Native.
- **Multi-language Support:** Expand accessibility for global audiences.